

Boundary Localization and Adjoining the Degree Derivation

Supplementary research note

September 8, 2026

1 Definitions, hypotheses, and scope

All vector spaces and tensor products are over $\mathbb{C}$. Write

$$\widehat{\mathfrak{sl}}_2 = (\mathfrak{sl}_2 \otimes \mathbb{C}[t, t^{-1}]) \oplus \mathbb{C}K, \quad U = U(\widehat{\mathfrak{sl}}_2)$$

for the affine algebra without the degree derivation. For $x \in \{e, h, f\}$ and $r \in \mathbb{Z}$, set $x_r = x \otimes t^r$. Our bracket conventions are

$$\begin{aligned} [e_r, f_s] &= h_{r+s} + r\delta_{r+s,0}K, & [h_r, h_s] &= 2r\delta_{r+s,0}K, \\ [h_r, e_s] &= 2e_{r+s}, & [h_r, f_s] &= -2f_{r+s}, \end{aligned} \tag{1}$$

where δ is the Kronecker delta, K is central, and $[e_r, e_s] = [f_r, f_s] = 0$. Adjoining the degree derivation gives

$$\widetilde{\mathfrak{sl}}_2 = \widehat{\mathfrak{sl}}_2 \rtimes \mathbb{C}d, \quad [d, x_r] = rx_r, \quad [d, K] = 0, \quad \widetilde{U} = U(\widetilde{\mathfrak{sl}}_2).$$

Fix $a, b \in \mathbb{Z}$ with $a + b \geq 0$, and put $L = a + b + 1 \geq 1$. Define

$$S_{a,b} = \text{span}_{\mathbb{C}}\{K, h_i \ (i \geq 0), e_i \ (i > a), f_j \ (j > b)\}.$$

The bracket relations show that this is a Lie subalgebra and that

$$[S_{a,b}, S_{a,b}] = \text{span}_{\mathbb{C}}\{e_i \ (i > a), f_j \ (j > b), h_k \ (k \geq L + 1)\}.$$

Choose $c, \lambda_0, \dots, \lambda_L \in \mathbb{C}$, and set $\lambda_i = 0$ for $i > L$. These parameters define a character $\chi : S_{a,b} \rightarrow \mathbb{C}$ by

$$\chi(K) = c, \quad \chi(h_i) = \lambda_i \quad (i \geq 0), \quad \chi(e_i) = \chi(f_j) = 0,$$

with the root indices restricted to those occurring in $S_{a,b}$. Set

$$M_{a,b}(\chi) = U \otimes_{U(S_{a,b})} \mathbb{C}_{\chi}, \quad \mathbb{C}_{\chi} = \mathbb{C}1_{\chi}, \quad s1_{\chi} = \chi(s)1_{\chi}, \quad u = 1 \otimes 1_{\chi}.$$

In particular, h_0 belongs to the inducing subalgebra. These conventions agree with [1, §2.3]. The main isomorphism requires $\lambda_L \neq 0$; the central parameter c is arbitrary.

For $F = f_n$, write

$$U_F = U[F^{-1}], \quad \mathcal{D}_F M = U_F \otimes_U M, \quad T_F M = \text{coker}(M \longrightarrow \mathcal{D}_F M).$$

The Ore localizations used here are justified in Section 4. When F acts injectively, we identify M with its image in $\mathcal{D}_F M$ and write $T_F M = \mathcal{D}_F M / M$. For $\widetilde{\mathfrak{sl}}_2$ -modules, use $\widetilde{U}_F$ in the same definitions. An expression $F^{-m}u + M$ denotes a coset; it does *not* assert that F^{-1} acts on the quotient module. A module V is called *smooth* if, for each $v \in V$, all sufficiently high modes e_r, h_r, f_r annihilate v .

Two distinct questions. Replacing f_1 by f_n may mean moving the inducing boundary along with n , or keeping the original module $M_{-1,1}$ fixed. The first interpretation admits a uniform isomorphism theorem; the second does not always do so. Likewise, adjoining $\mathbb{C}[d]$ requires a specified action satisfying $[d, x_r] = rx_r$. An unshifted tensor-product action does not in general satisfy this relation.

2 Localization at an arbitrary boundary

Theorem 2.1. Assume $a + b \geq 0$, $L = a + b + 1$, and $\lambda_L \neq 0$. Let $M = M_{a,b}(\chi)$ and $F = f_b$. Define a character on $S_{a+1,b-1}$ by

$$\chi^+(K) = c, \quad \chi^+(h_i) = \lambda_i + 2\delta_{i,0} \quad (i \geq 0),$$

with value zero on all root vectors in that subalgebra. Let v be the canonical inducing vector of $M_{a+1,b-1}(\chi^+)$. There is a unique $\widehat{\mathfrak{sl}}_2$ -module isomorphism

$$\Phi : M_{a+1,b-1}(\chi^+) \xrightarrow{\sim} T_{f_b} M_{a,b}(\chi), \quad \Phi(v) = F^{-1}u + M. \quad (2)$$

This holds for every $c \in \mathbb{C}$, including the critical level $c = -2$, and also allows $b \leq 0$.

2.1 PBW bases and well-definedness

Put $Y = e_{a+1}$ and $A = M_{a+1,b-1}(\chi^+)$. Let $\mathcal{X}$ consist of the finite ordered monomials

$$X = e(\alpha)h(\beta)f(\gamma), \quad \alpha_i \leq a, \quad \beta_i \leq -1, \quad \gamma_i \leq b-1.$$

Here α, β, γ are finite nondecreasing sequences, $e(\alpha) = e_{\alpha_1} \cdots e_{\alpha_{\ell(\alpha)}}$, and similarly for $h(\beta)$ and $f(\gamma)$. Empty products are allowed, and ℓ denotes the number of factors. Define

$$\rho(X) = 2\ell(\alpha) + \ell(\beta), \quad w_m = F^{-m}u + M, \quad B_{m,X} = F^{-m}Xu + M \quad (m \geq 1).$$

For M , choose a PBW order with F first, followed by the generators occurring in X , and then a basis of $S_{a,b}$. Thus $\{F^q Xu : q \geq 0, X \in \mathcal{X}\}$ is a basis of M , so F acts injectively. Clearing a common denominator and using this basis shows that $\{F^q Xu : q \in \mathbb{Z}, X \in \mathcal{X}\}$ is a basis of $D_F M$. Consequently,

$$\begin{aligned} \{B_{m,X} : m \geq 1, X \in \mathcal{X}\} & \text{ is a basis of } T_F M, \\ \{XY^p v : p \geq 0, X \in \mathcal{X}\} & \text{ is a basis of } A. \end{aligned} \quad (3)$$

For the source basis, the PBW order places Y after every factor of X and before the inducing subalgebra. These complements are used only as vector spaces; they need not be Lie subalgebras.

Write $\Delta = \text{ad } F$. The finite commutation formula (10) gives

$$\begin{aligned} h_i F^{-m} &= F^{-m} h_i + 2m F^{-m-1} f_{i+b}, \\ e_i F^{-m} &= F^{-m} e_i - m F^{-m-1} (h_{i+b} + i \delta_{i+b,0} K) \\ &\quad - m(m+1) F^{-m-2} f_{i+2b}. \end{aligned}$$

It follows that

$$h_i w_1 = (\lambda_i + 2\delta_{i,0}) w_1 \quad (i \geq 0), \quad K w_1 = c w_1 :$$

if $i > 0$, then $f_{i+b} u = 0$, whereas for $i = 0$ the correction term is $2F^{-1}u$. Also, $f_j w_1 = 0$ for $j > b$, and at the boundary $f_b w_1 = u + M = 0$. If $i \geq a + 2$, then $i + b \geq L + 1 > 0$, whence

$$e_i u = h_{i+b} u = f_{i+2b} u = 0, \quad \delta_{i+b,0} = 0.$$

Thus $e_i w_1 = 0$. This verifies every inducing relation for $S_{a+1,b-1}$.

Extend χ^+ to the algebra character $\chi_U^+ : U(S_{a+1,b-1}) \rightarrow \mathbb{C}$. By taking finite products and linear combinations of the verified relations, we obtain $sw_1 = \chi_U^+(s)w_1$ for every $s \in U(S_{a+1,b-1})$. Hence the linear map

$$U \otimes_{\mathbb{C}} \mathbb{C}_{\chi^+} \longrightarrow T_F M, \quad x \otimes \zeta 1_{\chi^+} \longmapsto \zeta x w_1$$

annihilates every balancing relation, since

$$xs \otimes \zeta 1_{\chi^+} - x \otimes s(\zeta 1_{\chi^+}) \longmapsto \zeta x (sw_1 - \chi_U^+(s)w_1) = 0.$$

It therefore descends to a well-defined map $\Phi : A \rightarrow T_F M$. The map commutes with left multiplication by every element of U , so it is a module homomorphism. Uniqueness follows from $A = Uv$. No use of $\lambda_L \neq 0$ has been made at this stage.

2.2 Explicit preimages of pure negative powers

Since $Yu = 0$, $[Y, F] = h_L$, and $f_{b+L}u = 0$, with no central term because $L > 0$, the preceding formulas yield

$$Yw_m = -m\lambda_L w_{m+1}, \quad Y^p w_1 = (-1)^p p! \lambda_L^p w_{p+1}.$$

Therefore

$$q_m = \frac{(-1)^{m-1}}{(m-1)! \lambda_L^{m-1}} Y^{m-1} v \quad (m \geq 1) \quad (4)$$

satisfies $\Phi(q_m) = w_m$. In particular, every $f_b^{-m}u + M$ already has an explicit preimage. This calculation does not assume surjectivity.

2.3 The surjectivity induction and injectivity

Assign weights to the generators of U by

$$\text{wt}(e_i) = 2, \quad \text{wt}(h_i) = 1, \quad \text{wt}(f_i) = \text{wt}(K) = 0.$$

Let $U_{\leq r}$ be the span of words of total weight at most r , with $U_{\leq r} = 0$ for $r < 0$. Every nonzero bracket of generators has weight at most the sum of their weights minus one, so PBW reordering does not increase weight. On each generator, Δ lowers weight by at least one whenever its value is nonzero. The Leibniz rule therefore gives

$$\Delta^j(X) \in U_{\leq \rho(X)-j}, \quad \Delta^j(X) = 0 \quad (j > \rho(X)).$$

Reorder $\Delta^j(X)u$ in the PBW basis of M . Inducing factors act by scalars or zero, which again cannot increase weight. Thus

$$\Delta^j(X)u = \sum_{q, X'} c_{j,q,X'} F^q X' u, \quad q \geq 0, \quad c_{j,q,X'} \neq 0 \implies \rho(X') \leq \rho(X) - j,$$

where the sum is finite. Substitute this into (10). The $j = 0$ term of Xw_m is exactly $B_{m,X}$. For $j \geq 1$, a term with $q \geq m + j$ vanishes in the quotient; otherwise it is a multiple of $B_{m+j-q,X'}$ with $\rho(X') < \rho(X)$. Hence

$$Xw_m = B_{m,X} + \text{a finite sum of terms of strictly smaller } \rho. \quad (5)$$

We prove by induction on $r \geq 0$ that $B_{m,X}$ lies in $\text{im } \Phi$ for every $m \geq 1$ and $\rho(X) \leq r$. *Base case* $r = 0$. Then X contains only f -factors and commutes with F . Thus

$$B_{m,X} = Xw_m = \Phi(Xq_m).$$

This includes every f -monomial, not only $X = 1$.

Induction step. Assume the assertion for $r - 1$. Fix any $m \geq 1$ and any X with $\rho(X) = r$. By (5),

$$\Phi(Xq_m) = B_{m,X} + \sum_{\nu=1}^s c_\nu B_{m_\nu, X_\nu}, \quad m_\nu \geq 1, \quad \rho(X_\nu) \leq r - 1.$$

For each ν , the induction hypothesis supplies $a_\nu \in A$ with $\Phi(a_\nu) = B_{m_\nu, X_\nu}$. Therefore

$$B_{m,X} = \Phi \left(Xq_m - \sum_{\nu=1}^s c_\nu a_\nu \right).$$

Each step involves only finitely many terms, and the nonnegative integer ρ strictly decreases in every correction term, so the recursion terminates. Crucially, the induction covers all denominator orders simultaneously; an increase in the denominator order causes no difficulty. By the target basis in (3), Φ is surjective.

For injectivity, (5) implies

$$\Phi(XY^p v) = (-1)^p p! \lambda_L^p B_{p+1, X} + \text{terms of strictly smaller } \rho.$$

In a nonzero finite linear combination of source basis vectors, consider the largest occurring value of ρ . Its terms map to distinct target basis vectors, with coefficients multiplied by the nonzero scalars $(-1)^p p! \lambda_L^p$. They cannot cancel. The image is therefore nonzero, proving injectivity and completing the proof of Theorem 2.1. Neither finite-dimensional filtration pieces nor simplicity are needed.

2.4 Special cases, iteration, and the fixed-module obstruction

Taking $a = -1$ and $b = n \geq 1$ gives the direct generalization

$$M_{0,n-1}(\chi^+) \cong T_{f_n} M_{-1,n}(\chi), \quad \lambda_n \neq 0, \quad (6)$$

with

$$q_m = \frac{(-1)^{m-1}}{(m-1)! \lambda_n^{m-1}} e_0^{m-1} v.$$

For successive boundary moves, put $N_0 = M_{a,b}(\chi)$ and $N_{r+1} = T_{f_{b-r}} N_r$. Then, for every finite $r \geq 0$,

$$N_r \cong M_{a+r, b-r}(\chi^{(r)}), \quad \chi^{(r)}(h_0) = \lambda_0 + 2r,$$

with all other character values unchanged. Indeed, $a + b$, L , and the nonzero value λ_L are preserved at each step, so the theorem applies repeatedly. This is a statement about finitely many successive localization quotients of changing modules, not about infinite iteration or simultaneous localization.

The situation differs if the original module $M = M_{-1,1}(\chi)$ is kept fixed.

If $n \geq 2$, then $F = f_n$ annihilates u . For any Xu , choose $N \geq 1$ with $(\text{ad } F)^N(X) = 0$. In the expansion of $F^N Xu$, each term with $j < N$ contains a positive power of F acting on u , while the term with $j = N$ is zero. Thus F acts locally nilpotently, and $\mathcal{D}_F M = T_F M = 0$.

If $n \leq 0$, then $F = f_n$ belongs to the PBW complement. Placing it first shows that it acts injectively and that the quotient is nonzero. However,

$$h_1(F^{-1}u + M) = \lambda_1(F^{-1}u + M) + 2(F^{-2}f_{n+1}u + M).$$

Since $n + 1 \leq 1$ and $n + 1 \neq n$, the correction term is a nonzero PBW basis vector, linearly independent of $F^{-1}u + M$. Hence $F^{-1}u + M$ is not an h_1 -eigenvector and cannot be the image of the canonical inducing vector of a standard module of the above form. This rules out only the specified cyclic map: it neither classifies the quotient nor excludes an isomorphism using a different cyclic vector.

Finally, if $\lambda_L = 0$, the cyclic map in Theorem 2.1 remains well-defined, but $\Phi(Yv) = 0$ whereas $Yv \neq 0$. Thus that map is not an isomorphism.

3 Adjoining the derivation: the correct tensor action

For any $\widehat{\mathfrak{sl}}_2$ -module M , define

$$\mathcal{I}(M) = \widetilde{U} \otimes_U M.$$

Ordering d before a basis of $\widehat{\mathfrak{sl}}_2$ gives the right U -module decomposition

$$\widetilde{U} = \bigoplus_{k \geq 0} d^k U, \quad \mathcal{I}(M) \cong \mathbb{C}[z] \otimes M, \quad P(z) \otimes m \longleftrightarrow P(d) \otimes_U m.$$

Here z is a formal polynomial variable, distinguished from the derivation d . Flipping the vector-space factors gives the proposed space $M \otimes \mathbb{C}[d]$; we keep the polynomial factor on the left. The PBW decomposition also shows that $\mathcal{I}$ is exact: kernels and images after induction are computed coefficientwise.

Lemma 3.1. *Under this identification, the induced action is*

$$\begin{aligned} d(P(z) \otimes m) &= zP(z) \otimes m, \\ x_r(P(z) \otimes m) &= P(z - r) \otimes x_r m, \\ K(P(z) \otimes m) &= P(z) \otimes Km. \end{aligned} \tag{7}$$

These formulas define a $\widetilde{\mathfrak{sl}}_2$ -module and realize $\mathcal{I}(M)$.

Proof. The identity $x_r d = (d - r)x_r$ gives $x_r P(d) = P(d - r)x_r$ by induction, so the formulas follow from the induced action. They may also be checked directly. For homogeneous modes,

$$[x_r, y_s](P(z) \otimes m) = P(z - r - s) \otimes [x_r, y_s]m.$$

The noncentral bracket term has degree $r + s$, and a central term can occur only when $r + s = 0$, exactly as required. Moreover,

$$\begin{aligned} [d, x_r](P(z) \otimes m) &= [zP(z - r) - (z - r)P(z - r)] \otimes x_r m \\ &= rP(z - r) \otimes x_r m. \end{aligned}$$

Together with centrality of K , this verifies all relations. $\square$

The unshifted action $x_r(P \otimes m) = P \otimes x_r m$ would give $[d, x_r] = 0$, contradicting the required relation whenever a nonzero-degree mode acts nontrivially. Nor can one simply extend the one-dimensional inducing character to include d : any such extension, denoted $\widehat{\chi}$, would satisfy

$$0 = [\widehat{\chi}(d), \widehat{\chi}(h_L)] = \widehat{\chi}([d, h_L]) = L\lambda_L \neq 0.$$

Thus we retain $S_{a,b}$, without d , as the inducing subalgebra. Transitivity of induction gives

$$\widetilde{M}_{a,b}(\chi) := \widetilde{U} \otimes_{U(S_{a,b})} \mathbb{C}_\chi \cong \mathcal{I}(M_{a,b}(\chi)). \tag{8}$$

4 Localization commutes with adjoining the derivation

This section does not require M to be a particular induced module. Fix $n \in \mathbb{Z}$ and $F = f_n$. For $\Delta = \text{ad } F$, the brackets give

$$\begin{aligned} \Delta(f_r) &= \Delta(K) = 0, & \Delta(h_r) &= 2f_{r+n}, & \Delta(d) &= -nF, \\ \Delta(e_r) &= -(h_{r+n} + r\delta_{r+n,0}K), & \Delta^2(e_r) &= -2f_{r+2n}, & \Delta^3(e_r) &= 0. \end{aligned}$$

The Leibniz rule implies that Δ is locally nilpotent on both U and $\tilde{U}$. Their PBW associated graded algebras are domains, so F is regular on both sides in each algebra. If $\Delta^{t+1}(s) = 0$, then for $N \geq t$,

$$F^N s = \sum_{j=0}^t \binom{N}{j} \Delta^j(s) F^{N-j}, \quad s F^N = \sum_{j=0}^t (-1)^j \binom{N}{j} F^{N-j} \Delta^j(s).$$

Taking $N \geq m+t$ gives the required right or left factor F^m , proving both Ore conditions. Module localization can be represented by fractions, with

$$F^{-m}v = F^{-\ell}w \iff F^{k-m}v = F^{k-\ell}w \text{ for some } k \geq \max\{m, \ell\}.$$

Thus finite sums admit a common denominator, and the kernel of $M \rightarrow \mathcal{D}_F M$ consists exactly of vectors annihilated by some power of F .

Theorem 4.1. *Suppose $F = f_n$ acts injectively on the $\widehat{\mathfrak{sl}}_2$ -module M . There are natural $\tilde{\mathfrak{sl}}_2$ -module isomorphisms*

$$\alpha_M : \mathcal{D}_F \mathcal{I}(M) \xrightarrow{\sim} \mathcal{I}(\mathcal{D}_F M), \quad \bar{\alpha}_M : T_F \mathcal{I}(M) \xrightarrow{\sim} \mathcal{I}(T_F M).$$

The first isomorphism and its inverse are given by

$$\begin{aligned} \alpha_M(F^{-m}(P(z) \otimes u)) &= P(z + mn) \otimes F^{-m}u, \\ \alpha_M^{-1}(P(z) \otimes F^{-m}u) &= F^{-m}(P(z - mn) \otimes u), \end{aligned} \tag{9}$$

where $m \geq 0$, $P \in \mathbb{C}[z]$, and $u \in M$. The second isomorphism is obtained by passing to quotients.

Proof. Injective action and construction. On $\mathcal{I}(M)$, the action of F is

$$P(z) \otimes u \longmapsto P(z - n) \otimes Fu.$$

Polynomial translation is invertible and F is injective on M ; hence F is injective on $\mathcal{I}(M)$. On $\mathcal{I}(\mathcal{D}_F M)$, it is invertible, with

$$F^{-1}(P(z) \otimes v) = P(z + n) \otimes F^{-1}v.$$

Iteration gives the positive shift $+mn$ in (9). The basic fraction relation is respected because

$$P(z - n + (m+1)n) \otimes F^{-m-1}Fu = P(z + mn) \otimes F^{-m}u.$$

Together with common denominators, this proves that α_M is well-defined.

Module compatibility. Local nilpotence gives the finite commutation identity

$$s F^{-m} = \sum_{j \geq 0} \binom{m+j-1}{j} F^{-m-j} \Delta^j(s) \quad (m \geq 1). \tag{10}$$

For $m = 1$, repeatedly substitute $s F^{-1} = F^{-1}s + F^{-1}\Delta(s)F^{-1}$. The general case follows by induction using the binomial summation identity obtained from Pascal's identity. Each nonzero $\Delta^j(x_r)$ is homogeneous of degree $r + jn$; a possible central term has degree zero. Therefore, for $m \geq 1$,

$$\begin{aligned} \alpha_M(x_r F^{-m}(P(z) \otimes u)) &= \sum_{j \geq 0} \binom{m+j-1}{j} P(z - (r + jn) + (m+j)n) \otimes F^{-m-j} \Delta^j(x_r)u \\ &= P(z + mn - r) \otimes x_r F^{-m}u \\ &= x_r \alpha_M(F^{-m}(P(z) \otimes u)). \end{aligned}$$

For $m = 0$, the assertion is immediate without commuting a denominator. Compatibility with K follows directly. For the derivation, $[d, F^{-m}] = -mnF^{-m}$, so

$$\begin{aligned}\alpha_M(dF^{-m}(P(z) \otimes u)) &= \alpha_M(F^{-m}((z - mn)P(z) \otimes u)) \\ &= zP(z + mn) \otimes F^{-m}u \\ &= d\alpha_M(F^{-m}(P(z) \otimes u)).\end{aligned}$$

Bijectivity and passage to quotients. The second formula in (9) supplies a preimage for every elementary tensor, proving surjectivity. If $\alpha_M(F^{-m}w) = 0$, apply F^m in the target: the image of w in $\mathcal{I}(\mathcal{D}_F M)$ is zero. Since $M \hookrightarrow \mathcal{D}_F M$ is injective and $\mathcal{I}$ is exact, $\mathcal{I}(M) \hookrightarrow \mathcal{I}(\mathcal{D}_F M)$ is injective. Thus $w = 0$, proving injectivity and the stated inverse formula.

On elements with denominator exponent zero, α_M is the induced inclusion. Applying $\mathcal{I}$ to

$$0 \longrightarrow M \longrightarrow \mathcal{D}_F M \longrightarrow T_F M \longrightarrow 0$$

therefore gives the quotient isomorphism. Finally, for a module map $g : M \rightarrow N$, the localized and induced maps send

$$F^{-m}u \longmapsto F^{-m}g(u), \quad P \otimes u \longmapsto P \otimes g(u).$$

Substitution in (9) proves naturality. $\square$

Corollary 4.2. *Under the hypotheses of Theorem 4.1, a $\widehat{\mathfrak{sl}}_2$ -module isomorphism $\Phi : A \xrightarrow{\sim} T_F M$ lifts to*

$$\tilde{\Phi} = \bar{\alpha}_M^{-1} \circ \mathcal{I}(\Phi) : \mathcal{I}(A) \xrightarrow{\sim} T_F \mathcal{I}(M).$$

If $\Phi(a) = F^{-m}u + M$, then

$$\tilde{\Phi}(P(z) \otimes a) = F^{-m}(P(z - mn) \otimes u) + \mathcal{I}(M). \quad (11)$$

In particular, if $\Phi(q_m) = F^{-m}u + M$, then

$$\boxed{\tilde{\Phi}^{-1}(F^{-m}(P(z) \otimes u) + \mathcal{I}(M)) = P(z + mn) \otimes q_m.} \quad (12)$$

The induced map $\mathcal{I}(\Phi)$ itself is indeed $\text{id}_{\mathbb{C}[z]} \otimes \Phi$. However, identifying its target with $T_F \mathcal{I}(M)$ requires the translation in (9). When $n \neq 0$, replacing $P \otimes (F^{-m}u + M)$ by $F^{-m}(P \otimes u) + \mathcal{I}(M)$ without this translation is generally incompatible with d . When $n = 0$, the translations disappear.

5 The combined isomorphism

Apply Corollary 4.2 to Theorem 2.1 and use (8).

Corollary 5.1. *If $a + b \geq 0$, $L = a + b + 1$, and $\lambda_L \neq 0$, then for every central parameter c ,*

$$\boxed{\widetilde{M}_{a+1,b-1}(\chi^+) \xrightarrow[\tilde{\Phi}]{\sim} T_{f_b} \widetilde{M}_{a,b}(\chi).} \quad (13)$$

The map sends $\tilde{v} = 1 \otimes v$ to $f_b^{-1}\tilde{u} + \widetilde{M}_{a,b}(\chi)$, where $\tilde{u} = 1 \otimes u$. For q_m from (4),

$$\begin{aligned}\tilde{\Phi}^{-1}(f_b^{-m}(P(z) \otimes u) + \widetilde{M}_{a,b}(\chi)) \\ = P(z + mb) \otimes q_m.\end{aligned} \quad (14)$$

In particular, for $a = -1$ and $b = n \geq 1$,

$$\widetilde{M}_{0,n-1}(\chi^+) \cong T_{f_n} \widetilde{M}_{-1,n}(\chi), \quad \lambda_n \neq 0.$$

This completes the proposed procedure. Well-definedness, the surjectivity induction, and injectivity first establish Φ without d . Exact induction gives $\mathcal{I}(\Phi)$, and $\bar{\alpha}_M^{-1}$ identifies its target with the localization quotient after adjoining d . No new surjectivity induction on the polynomial degree is needed. Likewise, each finite sequence of boundary moves lifts step by step.

6 Smoothness and limitations

If M is smooth, the finitely many coefficients m_i in $\sum_i P_i(z) \otimes m_i$ have a common bound above which all modes annihilate them. By (7), the same bound works for this tensor. Thus $\mathcal{I}(M)$ is smooth.

For an element $F^{-m}u$ of the localization, with $m \geq 1$, the finite commutation formula gives

$$\begin{aligned} h_r F^{-m}u &= F^{-m}h_ru + 2mF^{-m-1}f_{r+n}u, \\ e_r F^{-m}u &= F^{-m}e_ru - mF^{-m-1}(h_{r+n} + r\delta_{r+n,0}K)u \\ &\quad - m(m+1)F^{-m-2}f_{r+2n}u, \end{aligned}$$

while f_r commutes with F . Choose r sufficiently large that $r, r+n, r+2n$ exceed the annihilation bound for u and $r+n \neq 0$. All terms then vanish. Common denominators and finite sums show that $\mathcal{D}_F M$, and hence $T_F M$, are smooth. This argument also covers negative integers n .

The modules $M_{a,b}(\chi)$ themselves are smooth: commuting a sufficiently high mode through a fixed finite PBW word introduces only finitely many index shifts, and the resulting high modes annihilate the inducing vector.

An isomorphism statement does not by itself imply simplicity. In general, $\mathcal{I}$ does not preserve simple modules: the trivial one-dimensional $\widehat{\mathfrak{sl}}_2$ -module $\mathbb{C}$ is simple, but $\mathcal{I}(\mathbb{C}) = \mathbb{C}[z]$ has the nonzero proper submodule $z\mathbb{C}[z]$, with all loop modes acting as zero and d acting by multiplication by z . This example only shows that induction does not preserve simplicity in general; it does not satisfy the injectivity hypothesis on F .

For the noncritical induced modules considered here, [1, §4.1] uses an additional Casimir central reduction to construct simple modules. The isomorphisms in this note alone do not establish simplicity of the entire induced module with d .

A Scope and record of the Danus trial

The trial used the public Codex release of Danus, `v0.1.0-codex`, through a bounded native Windows adaptation. It retained the upstream deterministic prechecks, verification entry point, candidate-submission gate, and fact graph, while replacing the Linux daemon launch mechanism with local read-only Codex calls. No unsandboxed launch script was run, no additional paid strategic consultation was enabled, and no global agent configuration was changed.

Separate workers submitted candidates for the boundary isomorphism and for compatibility with the derivation. Each review used an independent cold-start verifier. The boundary candidate was accepted on its first submission, with fact identifier `d6bf6fef62dc8971`. The derivation candidate was initially rejected because its theorem statement did not fully define the ambient algebras and functor. After these definitions were supplied, it was accepted with fact identifier `b388fabb701946d3`. The second review confirmed consistency of the positive shift $+mn$, inverse shift $-mn$, and derivation correction.

The two English proof candidates in the fact graph retain their submitted statements, proofs, and verification results. The present note is an edited English version of the integrated Chinese note: it uses unified notation and combines the results functorially in Corollary 5.1. Neither integrated version has undergone a separate whole-paper verifier review. All 67 offline tests for the adaptation and relevant upstream modules passed; dependency deprecation warnings and nonblocking heuristic symbol-recognition warnings were also recorded. These software tests are not mathematical proofs.

The Danus verifier is an LLM, not Lean or Coq; acceptance does not constitute a formal proof certificate [2]. The arguments remain subject to author and advisor review. A complete classification of nonboundary quotients of the fixed original module, and further simplicity questions, lie outside the results established here.

References

- [1] V. Futorny, X. Guo, Y. Xue and K. Zhao, *Smooth representations of affine Kac–Moody algebras*, arXiv:2404.03855v2 (2024), §2.3 and §4. <https://arxiv.org/abs/2404.03855v2>.
- [2] FrenzyMath, *Danus: Security and Trust Model*, v0.1.0-codex, §1–4. <https://github.com/frenzymath/Danus/blob/v0.1.0-codex/docs/security-and-trust.md>.